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Class: CSE-4321-001

HW 5:

Exercise Section 7.2.3, Problem 1 (Graph II and Graph III only).

HW5

7.2.3 problem1

- Below are four graphs, each of which is defined by the sets of nodes, initial nodes, final nodes, edges, and defs and uses. Each graph also contains some test paths. Answer the following questions about each graph.

Graph II.

$N = \{1, 2, 3, 4, 5, 6\}$

$N_0 = \{1\}$

$N_f = \{6\}$

$E = \{(1, 2), (2, 3), (2, 6), (3, 4), (3, 5), (4, 5), (5, 2)\}$

$def(1) = def(3) = use(3) = use(6) = \{x\}$

// Assume the use of x in 3 precedes the def

Test Paths:

$t1 = [1, 2, 6]$

$t2 = [1, 2, 3, 4, 5, 2, 3, 5, 2, 6]$

$t3 = [1, 2, 3, 5, 2, 3, 4, 5, 2, 6]$

$t4 = [1, 2, 3, 5, 2, 6]$

Graph III.

$N = \{1, 2, 3, 4, 5, 6\}$

$N_0 = \{1\}$

$N_f = \{6\}$

$E = \{(1, 2), (2, 3), (3, 4), (3, 5), (4, 5), (5, 2), (2, 6)\}$

$def(1) = def(4) = use(3) = use(5) = use(6) = \{x\}$

Test Paths:

$t_1 = [1, 2, 3, 5, 2, 6]$

$t_2 = [1, 2, 3, 4, 5, 2, 6]$

- Draw the graph.

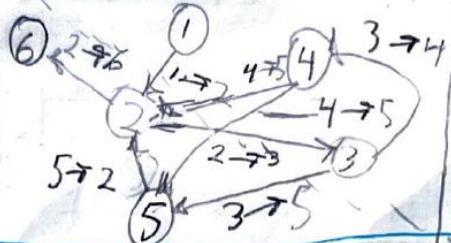
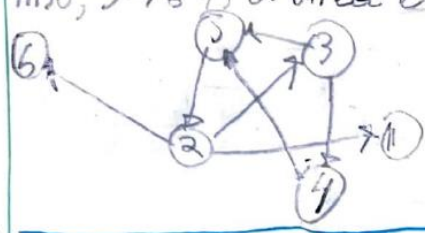
G = Graph II or III

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G II
A

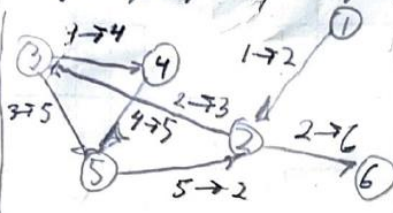
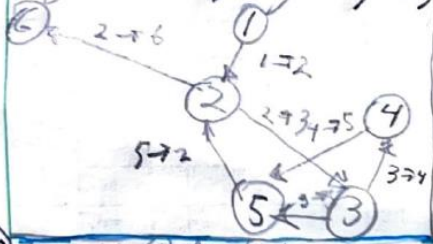
Nodes: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 2$ (loop), $3 \rightarrow 6$
 edges: $(1,2), (2,3), (2,6), (3,4), (3,5), (4,5), (5,2)$
 graph has a loop: $2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 2$
 Also, $3 \rightarrow 6$ is a direct edge

D.C.



G III
A

Nodes: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 2$ (loop), $2 \rightarrow 6$
 edges: $(1,2), (2,3), (3,4), (3,5), (4,5), (5,2), (2,6)$



G III B

def X @ Node 1, 2, 3, 4, 5, 6

- (b) List all of the du-paths with respect to x . (Note: Include all-du-paths, even those that are subpaths of some other du-path).

611B def X: @ Node 1
 use X: @ Node 3 & 6
 Find all def-clear paths
 ✓ [1, 2, 3] ← use 3
 [1, 2, 3, 4, 5, 2, 3] → Loop back to use 3
 ✓ [1, 2, 3] → use 6
 [1, 2, 3, 6] → use 6
 ✓ [1, 2, 3, 4, 5, 2, 6] → use 6
 [1, 2, 3, 4, 5, 2, 3, 6] → looped version to use 6

1, 2, 3 ← use 3
 1, 2, 6 ← use 6
 3 ← Node 3
 3, 4, 5, 2, 6 ← use 6

611B def X: @ Nodes 1 & 4
 use X: @ Nodes 3, 5, 6
 def (4):
 ① ✓ [1, 2, 3] → use 3
 [1, 2, 3, 4, 5] → use 5
 ② ✓ [1, 2, 3, 5] → use 5
 ③ ✓ [1, 2, 6] → use 6
 def (4)
 ⑤ ✓ [4, 5] → use (5)
 ⑥ ✓ [4, 5, 2, 6] → use (6)
 All du-paths:
 [1, 2, 3] [4, 5]
 [1, 2, 3, 4, 5] [4, 5, 2, 6]
 [1, 2, 3, 5]
 [1, 2, 6]

No?
 [1, 2, 3, 5, 6] ✓ ④
 [1, 2, 3, 4, 5, 6]

= double check

- (c) Determine which du-paths each test path tours. Write them in a table with test paths in the first column and the du-paths they cover in the second column. For this part of the exercise, you should consider both direct touring and sidetrips.

G116 Test Path

t1 = [1, 2, 6]
 t2 = [1, 2, 3, 4, 5, 2, 3, 5, 2, 6]
 t3 = [1, 2, 3, 5, 2, 3, 4, 5, 2, 6]
 t4 = [1, 2, 3, 5, 2, 6]

Test Path Covered?

[1, 2, 6]
 [1, 2, 3] [1, 2, 3, 4, 5, 2, 3] [1, 2, 3, 4, 5, 6]
 [1, 2, 3] for T3
 [1, 2, 3], [1, 2, 3, 4, 5, 2, 6] SKIPS 4

G1110

T1 = [1, 2, 3, 5, 2, 6] ✓ [1, 2, 3] [1, 2, 3, 5] [1, 2, 6] ✓
 T2 = [1, 2, 3, 4, 5, 2, 6] ✓ [1, 2, 3, 4, 5] [1, 2, 6] ✓
 NO PATHS STARTS AT Node 4 so [4, 5] ✓ [4, 5, 2, 6] ✓ are not covered

G11D

Sum. of 1 Node 1. Any test path starting at Node 1

(d) List a minimal test set that satisfies all defs coverage with respect to x. (Direct tours only.) Use the given test paths.

NO PATHS STARTS AT Node 7 so [4, 5] [4, 5, 2, 6]

G11D

only def 1 Node 1. Any test path starting at Node 1 satisfies this.
 any 1 of (t1, t2, t3, t4) works. Min. Set {t1} (part 15 t3 & t4)

G111D

Defs @ Nodes 1 & 4
 only test paths starting @ 4 are available. No test path covers path (4)
 so... NO valid test using given paths satisfies all defs. coverage. I think min set is t2 & t3/t4

G11E

use X @ 3 & 6
 T: # covers use 6 (1, 2, 6)

(e) List a minimal test set that satisfies all uses coverage with respect to x. (Direct tours only.) Use the given test paths.

0-11 def's. Coverage 14 think min set is t2 & t3 & t4 & t5 & t6 & t7 & t8 & t9 & t10 & t11 & t12 & t13 & t14 & t15 & t16 & t17 & t18 & t19 & t20 & t21 & t22 & t23 & t24 & t25 & t26 & t27 & t28 & t29 & t30 & t31 & t32 & t33 & t34 & t35 & t36 & t37 & t38 & t39 & t40 & t41 & t42 & t43 & t44 & t45 & t46 & t47 & t48 & t49 & t50 & t51 & t52 & t53 & t54 & t55 & t56 & t57 & t58 & t59 & t60 & t61 & t62 & t63 & t64 & t65 & t66 & t67 & t68 & t69 & t70 & t71 & t72 & t73 & t74 & t75 & t76 & t77 & t78 & t79 & t80 & t81 & t82 & t83 & t84 & t85 & t86 & t87 & t88 & t89 & t90 & t91 & t92 & t93 & t94 & t95 & t96 & t97 & t98 & t99 & t100

611E

uses 0, 3, 4, 6
T: # covers use 6 (1, 2, 6)
1 3
2 3
3 3
4 3
hits all (1, 2, 3), (1, 2, 6)
definitely NOT T4 just t3

611E

uses 0, 3, 5, 6
check which path reaches those
T1: uses 3, 5, 6
T2: 3, 5, 6 Min set (t1) (or t2)

611E Need to cover all 6 du-paths from b

(f) List a minimal test set that satisfies all-du-paths coverage with respect to x. (Direct tours only.) Use the given test paths.

611E

Need to cover all 6 du-paths from b
From c
T: # covers
1 [1, 2, 6]
2 all du-paths
3 all du-paths
4 misses 4 or more p.
Minimal path set: (t2) or (t3)
we have 4 du-paths
t3 hits all 4
so min set t3
def

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611F

Need to cover 6 du-paths from c
t1 covers: [1, 2, 3], [1, 2, 6], [1, 2, 3, 5]
t2 covers: [1, 2, 3, 4, 5]
No test using only given path satisfies all-du-paths coverage
double check: actually
we have 6 du-paths + t1 covers (1, 2, 3), (1, 2, 3, 5, 2, 6)
t2 covers all 6
so min set = (t2)